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Eddy Viscosity Adaptation

Using the Continuous Adjoint Method

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1 Introduction

RANS simulations are widely used in vehicle aerodynamics because of their relatively low computational cost. Their predictions of boundary layers, flow separation and the wake depend on the turbulence model. DES resolves part of the unsteady turbulent structures at a substantially higher computational cost.

In this project, eddy viscosity adaptation (EVA) is used to bring a RANS velocity field closer to a time-averaged DES reference. The turbulent kinematic viscosity, ν_t, is adjusted using sensitivities calculated with the continuous adjoint method. The method is tested on the DrivAer Fastback.

2 Mathematical Formulation

2.1 Governing equations

We start from the incompressible Navier–Stokes equations:

$$\nabla \cdot \mathbf{u} = 0, \qquad \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p_s + \nu \nabla^2 \mathbf{u}, \tag{1}$$

where $\mathbf{u}$ is the instantaneous velocity, p_s is the static pressure, ρ is the constant density and ν is the molecular kinematic viscosity.

Using Reynolds decomposition, we write $\mathbf{u} = \mathbf{U} + \mathbf{u}'$ and take the time average. For a steady mean flow, this gives

$$\nabla \cdot \mathbf{U} = 0, \qquad (\mathbf{U} \cdot \nabla)\mathbf{U} = -\frac{1}{\rho}\nabla \overline{p}_s + \nu \nabla^2 \mathbf{U} - \nabla \cdot \overline{\mathbf{u}' \otimes \mathbf{u}'}. \tag{2}$$

The averaging introduces the unknown Reynolds stresses. We model them using the Boussinesq approximation and the mean strain-rate tensor $\mathbf{S}$:

$$-\overline{u_i' u_j'} = 2\nu_t S_{ij} - \frac{2}{3}k\delta_{ij}, \qquad S_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right). \tag{3}$$

Here, $\mathbf{S}$ describes deformation by the mean flow, ν_t is the turbulent kinematic viscosity, $k = \frac{1}{2}\overline{u_i' u_i'}$ is the turbulent kinetic energy, and δ_{ij} is the Kronecker delta. If we substitute this expression into the momentum equation and combine the isotropic term with the pressure:

$$p^* = \frac{\overline{p}_s}{\rho} + \frac{2}{3}k. \tag{4}$$

The equations then become

$$\nabla \cdot \mathbf{U} = 0, \tag{5}$$

$$(\mathbf{U} \cdot \nabla)\mathbf{U} + \nabla p^* - \nabla \cdot [2(\nu + \nu_t)\mathbf{S}] = \mathbf{0}. \tag{6}$$

These equations are discretised using the finite-volume method on the computational mesh covering the fluid domain. In EVA, ν_t is prescribed during each primal solution and then updated through optimisation.

2.2 Objective and constraints

The reference velocity is defined over a DES averaging interval T_{avg} as

$$\mathbf{U}_{\text{ref}}(\mathbf{x}) = \frac{1}{T_{\text{avg}}} \int_{t_a}^{t_a + T_{\text{avg}}} \mathbf{U}_{\text{DES}}(\mathbf{x}, t)\, \mathrm{d}t. \tag{7}$$

The objective is the volume-integrated squared velocity mismatch:

$$J(\mathbf{U}) = \frac{1}{2} \int_{\Omega} |\mathbf{U} - \mathbf{U}_{\text{ref}}|^2 \, \mathrm{d}\Omega. \tag{8}$$

Its finite-volume approximation is

$$J_h = \frac{1}{2} \sum_{c=1}^{N_c} V_c \, |\mathbf{U}_c - \mathbf{U}_{\mathrm{ref},c}|^2 \,, \tag{9}$$

where V_c is the volume of cell c and N_c is the number of cells.

Let $\mathbf{q} = (\mathbf{U}, p)$ denote the flow state and $\mathbf{R}(\mathbf{q}, \nu_t) = 0$ the governing equations with their boundary conditions. The optimisation problem is

$$\begin{aligned} \underset{\nu_t}{\text{minimise}} \quad & J\big(\mathbf{q}(\nu_t)\big), \\ \text{subject to} \quad & \mathbf{R}(\mathbf{q}, \nu_t) = \mathbf{0}, \\ & \nu_{t,\min} \le \nu_t(\mathbf{x}) \le \nu_{t,\max}. \end{aligned} \tag{10}$$

We therefore seek an eddy-viscosity field that brings the computed velocity as close as possible to the reference, while satisfying the flow equations and the prescribed viscosity bounds. These bounds restrict the possible solutions, but do not guarantee a unique optimum or an exact match with the reference field.

2.3 Continuous adjoint formulation

We use a gradient-based optimisation method, so we need the sensitivity of the objective to the eddy viscosity in each cell. Calculating it by perturbing each variable separately would require too many flow solutions.

Instead, we solve the adjoint equations. Using $\mathbf{q} = (\mathbf{U}, p^*)$ for the primal fields and $\boldsymbol{\lambda} = (\mathbf{U}_a, p_a)$ for the adjoint fields, these equations can be written in compact form as

$$\left(\frac{\partial \mathbf{R}}{\partial \mathbf{q}}\right)^* \boldsymbol{\lambda} = -\begin{pmatrix} \mathbf{U} - \mathbf{U}_{\mathrm{ref}} \\ 0 \end{pmatrix}. \tag{11}$$

Here, $\partial \mathbf{R}/\partial \mathbf{q}$ describes how the flow-equation residuals change when velocity and pressure change. The star indicates the corresponding adjoint operator, including its boundary conditions. The right-hand side comes from differentiating the objective. It depends on the velocity mismatch and has no direct pressure contribution.

We solve this system using the current primal solution. It gives the auxiliary fields $\mathbf{U}_a$ and p_a, which are then used to calculate the viscosity sensitivity:

$$g = \left(\frac{\partial \mathbf{R}}{\partial \nu_t}\right)^* \boldsymbol{\lambda}. \tag{12}$$

The meaning of g is expressed by

$$\delta J = \int_\Omega g(\mathbf{x}) \, \delta\nu_t(\mathbf{x}) \, \mathrm{d}\Omega. \tag{13}$$

Thus, g gives the first-order effect of a small viscosity change on the objective. The optimisation algorithm uses it to choose the next correction.

For this single objective, one adjoint solution provides sensitivities throughout the domain. The method is called continuous because the adjoint equations are derived before discretisation.

2.4 Update strategy

At cycle k, steepest descent proposes a correction proportional to $-g_k$. Including the viscosity bounds, a trial update can be written schematically as

$$\nu_t^{\mathrm{trial}}(\alpha) = \Pi_{\mathcal{B}} \left[\nu_t^{(k)} - \alpha \eta_k g_k \right], \tag{14}$$

where $\Pi_{\mathcal{B}}$ projects onto the admissible bounds, η_k sets the correction scale and α is the line-search multiplier.

For a feasible descent direction d_k, the Armijo condition requires

$$\widehat{J}\left(\nu_t^{(k)} + \alpha d_k\right) \leq \widehat{J}\left(\nu_t^{(k)}\right) + c_1 \alpha \left\langle \widehat{g}_k, d_k \right\rangle, \qquad 0 < c_1 < 1, \tag{15}$$

Here, k is the optimisation-cycle index, d_k is the proposed search direction and α scales the trial step. The function $\widehat{J}$ is the merit function evaluated by the line search, and $\widehat{g}_k$ is its gradient at the current design.

The left-hand side is the trial merit value, calculated after updating the eddy viscosity and solving the primal flow again. The inner product $\langle \widehat{g}_k, d_k \rangle$ is the directional derivative: it estimates how the merit function changes along d_k. For a descent direction, it is negative.

The parameter c_1 specifies the fraction of the predicted reduction required for acceptance. The condition therefore accepts a step only if it produces sufficient reduction. Otherwise, α is reduced and another trial is evaluated, up to the prescribed line-search iteration limit.

Each optimisation cycle consists of a primal solution, an adjoint solution, sensitivity evaluation and a trial update. The accuracy of the sensitivities and the reliability of the line search depend on the convergence of the underlying flow solutions.

3 Numerical Methodology

All simulations use an inlet velocity of $U_\infty = 16 m/s$

Drag is expressed as

$$C_D = \frac{F_D}{\frac{1}{2}\rho U_\infty^2 A_{\text{ref}}}, \tag{16}$$

using a reference area of $2.176\,\text{m}^2$

3.1 Initial configuration and mesh study

The initial configuration is the DrivAer Fastback with a smooth underbody, mirrors, rotating wheels and moving ground. Three meshes were compared against the experimental value $C_{D,\text{exp}} = 0.243$ reported by Heft et al. (2012).

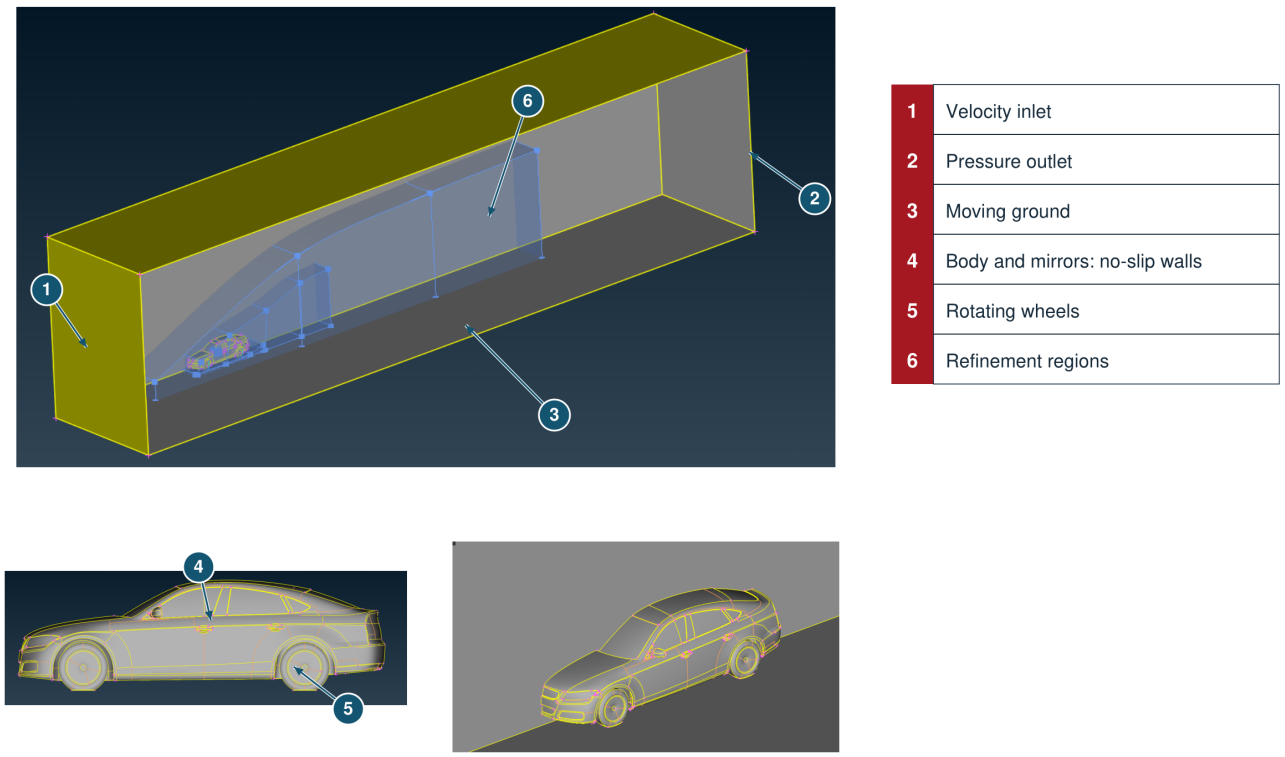


Figure 1: Initial DrivAer configuration and computational domain.

Table 1: Initial mesh study: comparison with $C_{D,\text{exp}} = 0.243$.

Cells (millions)	RANS C_D	DES $\overline{C_D}$	RANS difference (%)	DES difference (%)
5	0.2580	0.2495	6.17	2.67
7	0.2531	0.2502	4.16	2.96
11	0.2518	0.2485	3.62	2.26

RANS approaches the experimental drag with mesh refinement, while DES gives closer agreement on all three meshes. The mesh study was used to assess how well the simulations reproduce the measured drag and to identify a suitable mesh resolution for the subsequent EVA study. Based on this comparison, the 5M mesh was selected as a balance between agreement with the experiment and computational cost.

3.2 Configuration used for EVA

For the EVA study, the 5M cells configuration was simplified for both practical and numerical reasons. The mirrors and wheels were removed, the wheelhouses were closed, and the ground was kept stationary. This allowed us to focus on the main-body flow and wake with fewer interacting geometric features.

RANS and DES simulations were performed for this configuration as well, as the RANS solution provides the initial field, and the time-averaged DES velocity provides the target for EVA. For the corresponding simplified configuration, Heft et al. (2012) report $C_{D,\mathrm{exp}} = 0.124$.

Table 2: Simplified configuration: comparison with $C_{D,\mathrm{exp}} = 0.124$.

Simulation	Drag coefficient	Relative difference (%)
RANS	0.1316	6.14
DES	0.1287	3.86

Both predictions slightly overpredict the experimental drag, with DES giving closer agreement.

4 EVA Results and Discussion

4.1 Objective history

Figure 2 shows the objective's reduction over 38 optimisation cycles. The objective decreases from 11.5001 to 6.5236, giving a reduction of 43.27% relative to this state. Most of the decrease occurs during the first few displayed cycles, followed by a more gradual improvement.

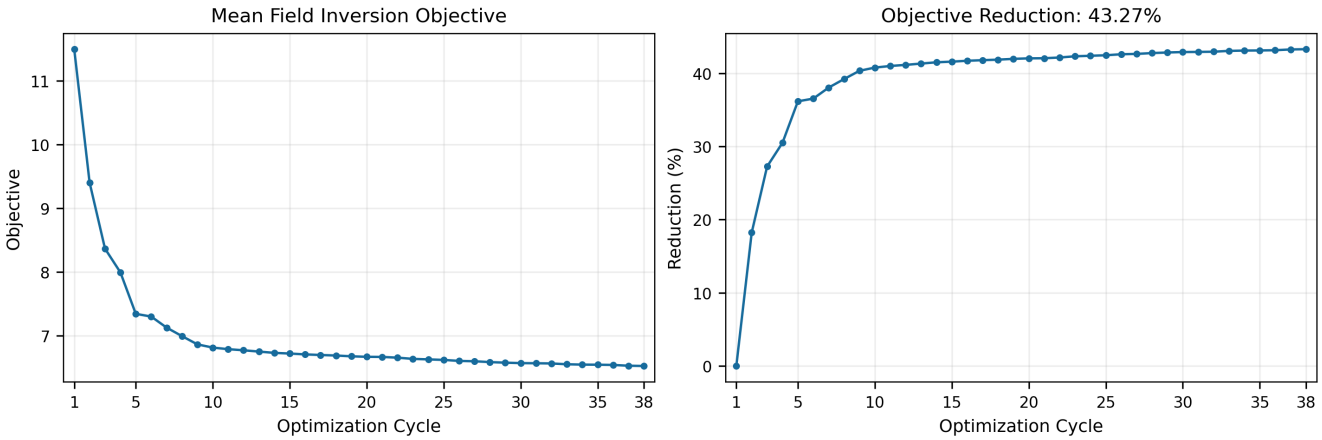


Figure 2: Mean field-inversion objective and percentage reduction relative to the first displayed state.

4.2 Velocity-field comparison

The following comparisons use the initial RANS field stored at time 0 and the final adapted mean field stored at time 38.

Figure 3 compares the velocity magnitude on the centre plane. All three fields show acceleration over the roof and a low-speed wake behind the vehicle.

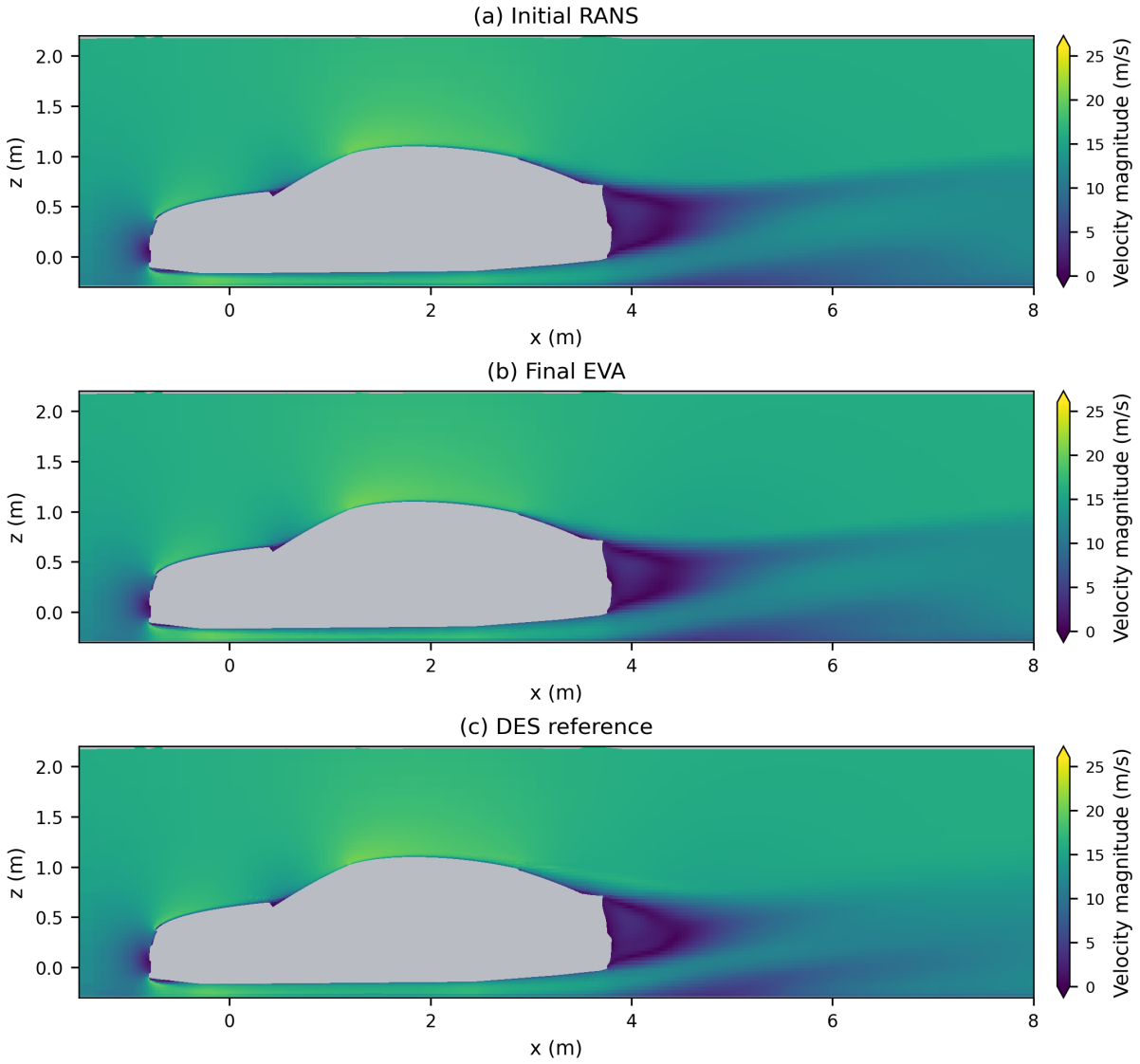


Figure 3: Velocity magnitude on the centre plane.

4.3 Wake direction

Figure 4 shows that downward motion extends lower into the wake in DES than in RANS. For example, near $(x, y, z) = (4.5, 0, 0.5)$ m, the vertical velocity is approximately

$$U_{Z,RANS} = 0.834 m/s,\ U_{Z,EVA} = 0.634 m/s,\ U_{Z,DES} = -1.641 m/s$$

for RANS, EVA and DES, respectively. At this position, EVA reduces the upward velocity but does not reproduce the downward motion predicted by DES.

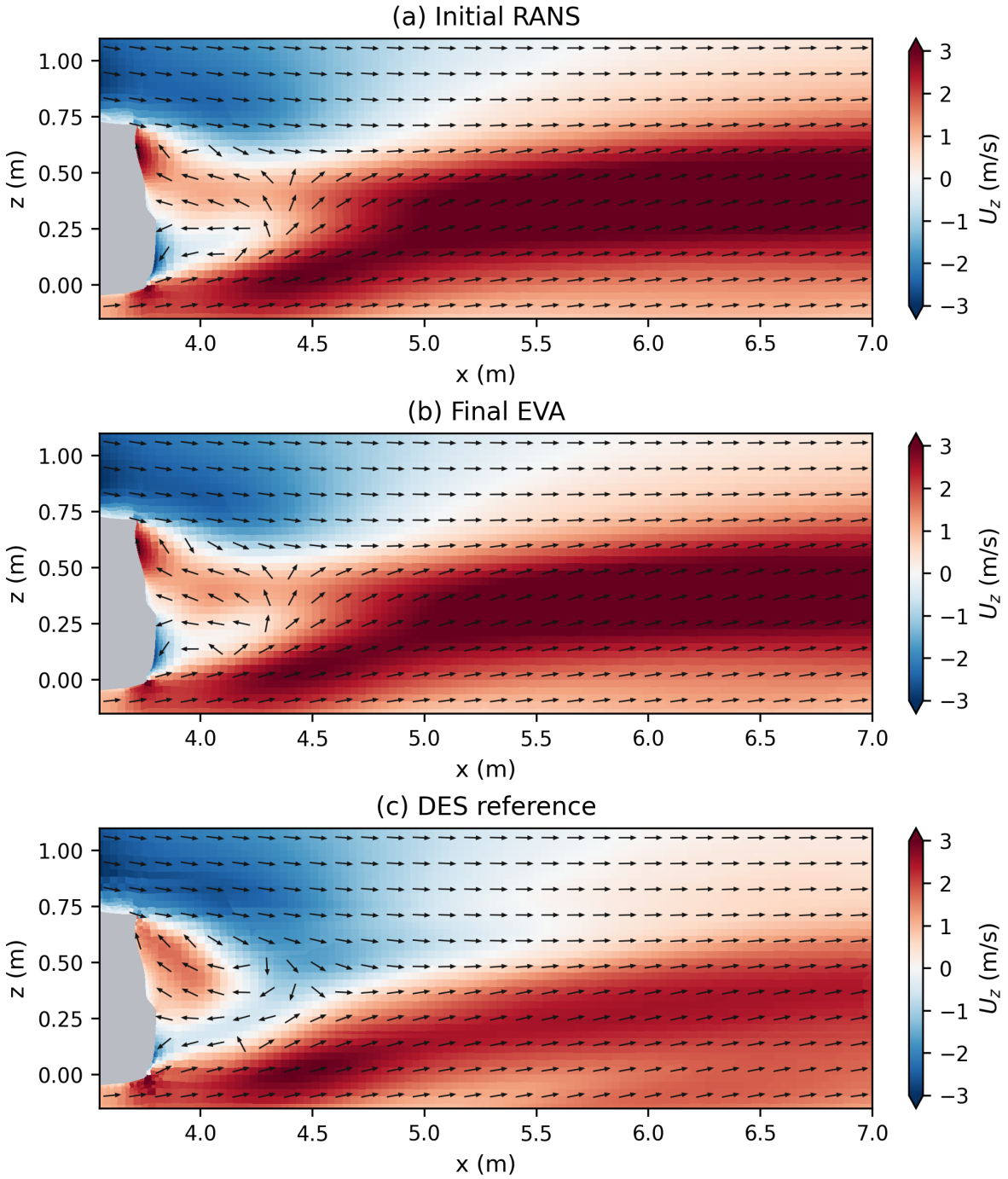


Figure 4: Wake velocity directions on the centre plane for initial RANS, final EVA and DES. Arrows show the projected (U_x, U_z) direction at identical positions, with equal lengths for readability. Colour shows U_z

Figure 5 quantifies this difference at three downstream positions. At $x = 4.5$ and 5 m, DES predicts negative U_z over a wider vertical region. EVA moves parts of the profiles towards DES, but the correction remains partial. At $x = 6$ m, the main difference is the stronger upward motion predicted by RANS and EVA over the intermediate wake heights.

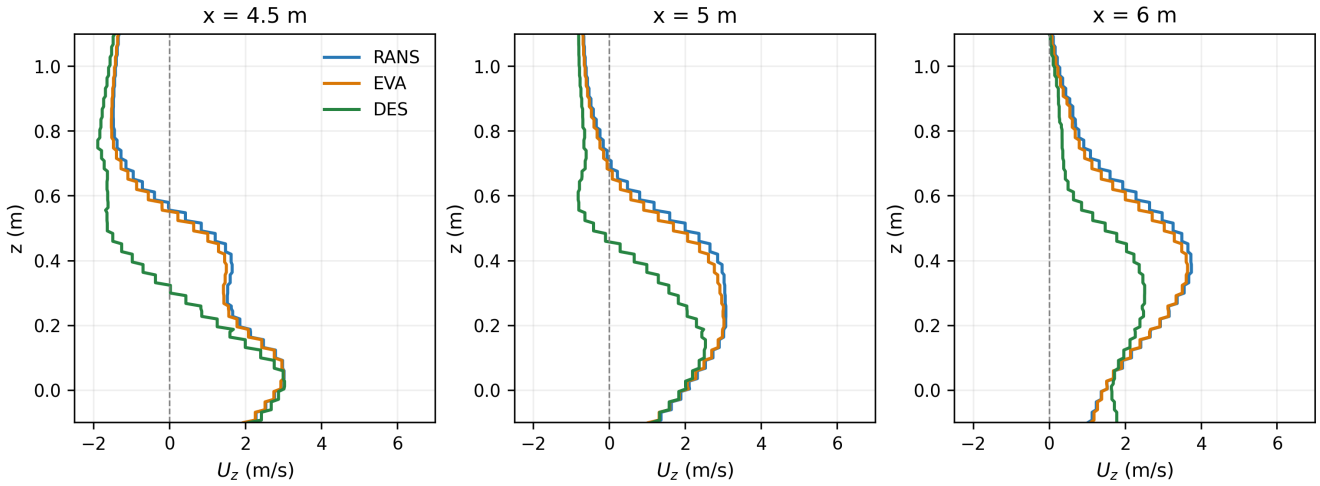


Figure 5: Vertical-velocity profiles on the centre plane at three downstream positions. Negative U_z indicates downward motion.

4.4 A local change in velocity direction

Figure 6 compares the velocity vectors in one cell of the lower near-wake region, at $(x, y, z) = (4.0009, 0.0831, 0.1787)$ m. This cell was selected to illustrate a case where both direction and total velocity mismatch improve. The angle to the DES vector is calculated using all three components:

$$\theta = \cos^{-1}\left(\frac{\mathbf{U} \cdot \mathbf{U}_{\text{ref}}}{|\mathbf{U}|\,|\mathbf{U}_{\text{ref}}|}\right). \tag{17}$$

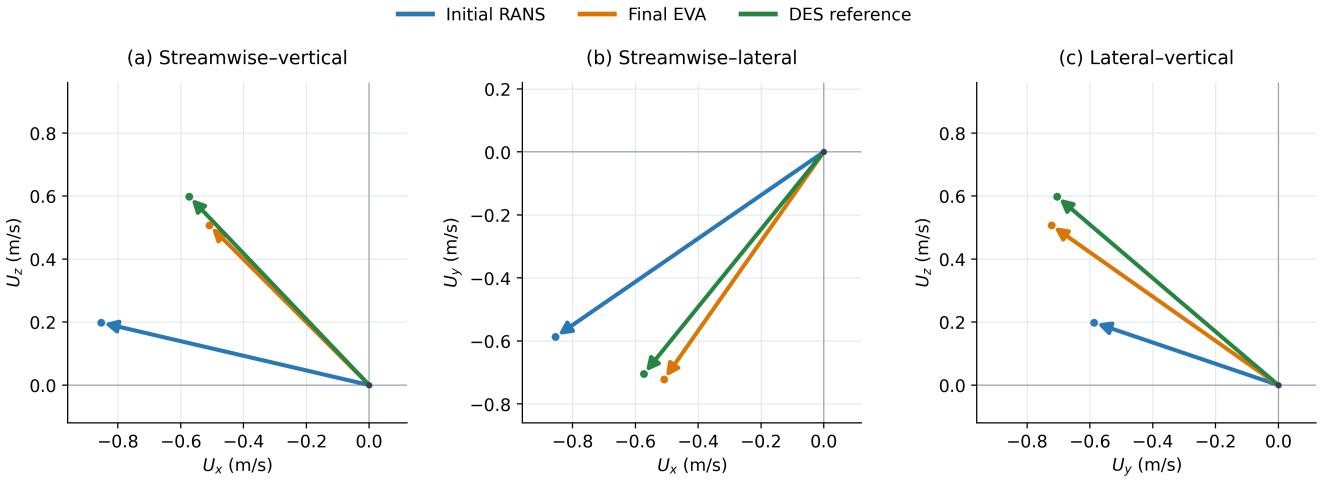


Figure 6: Velocity vectors in the selected cell, projected onto (a) the x–z, (b) the x–y, and (c) the y–z planes. Arrow lengths represent their corresponding projected values.

Table 3: Velocity components and mismatch in the selected cell. Velocities and errors are in m/s while θ is the angle to the DES vector.

Field	U_x	U_y	U_z	θ	e_U
RANS	-0.8538	-0.5867	0.1982	27.10°	0.5029
EVA	-0.5086	-0.7220	0.5071	4.86°	0.1129
DES	-0.5728	-0.7048	0.5983	0°	0

EVA reduces the angular mismatch from 27.10° to 4.86° and the vector error by 77.56% in this cell. The speeds remain similar: 1.055, 1.018 and 1.088 $\mathrm{m\,s^{-1}}$ for RANS, EVA and DES, respectively. The improvement mainly concerns direction, even though the speed itself becomes slightly less accurate.

This example shows how eddy-viscosity adaptation can change the direction of the predicted flow, rather than only its speed. All three vectors have $U_x < 0$ and $U_z > 0$, so this cell belongs to an upward-moving recirculation region.

4.5 Spatial distribution of the velocity error

The local velocity mismatch and its change are defined as

$$e_U = |\mathbf{U} - \mathbf{U}_{\mathrm{ref}}|, \qquad \Delta e_U = e_{U,\mathrm{final}} - e_{U,\mathrm{initial}}. \tag{18}$$

Negative Δe_U indicates improvement, while positive values indicate deterioration. This measure includes differences in both velocity magnitude and direction.

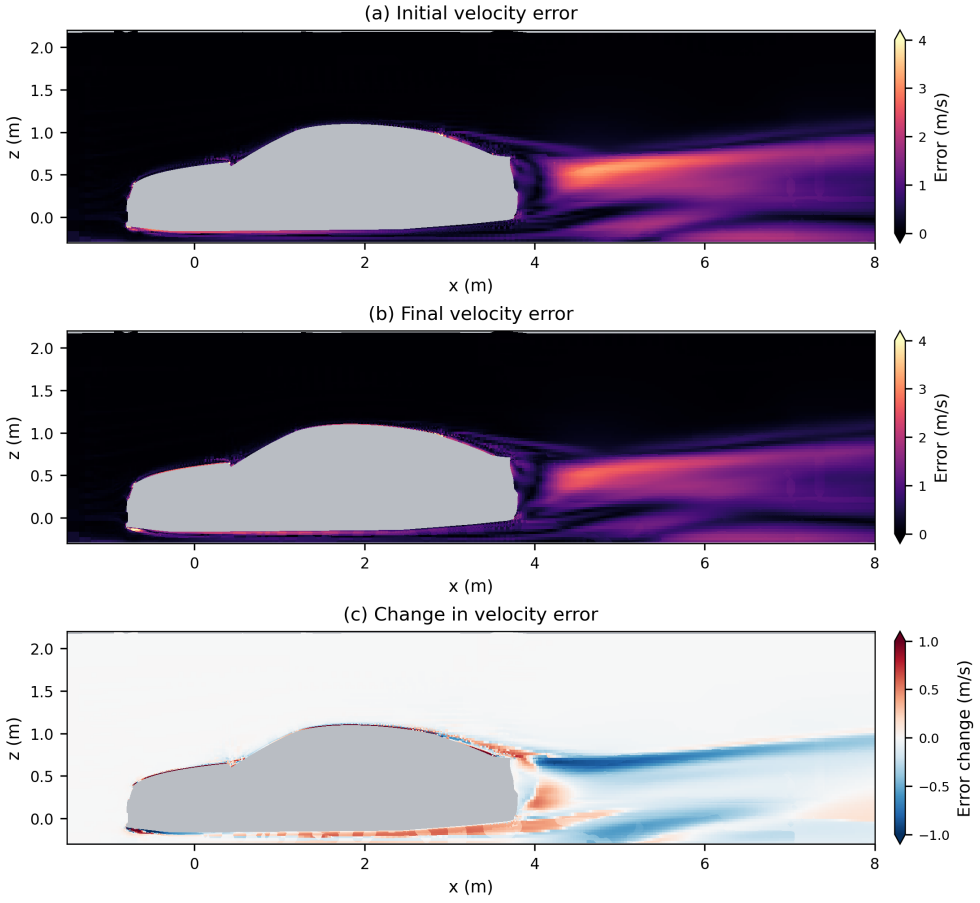


Figure 7: Centre-plane velocity error before and after adaptation, and its change.

On the centre plane, the error decreases along parts of the upper wake and in the downstream region close to the ground (Figure 7). Local increases remain near the rear body and along parts of the underbody. The adaptation therefore improves the overall agreement without reducing the error everywhere. The transverse section at $x = 4.5\,\mathrm{m}$ shows a similarly non-uniform response (Figure 8). Several high-error regions become weaker, particularly in the upper and lateral parts of the wake, while some neighbouring regions show an increase.

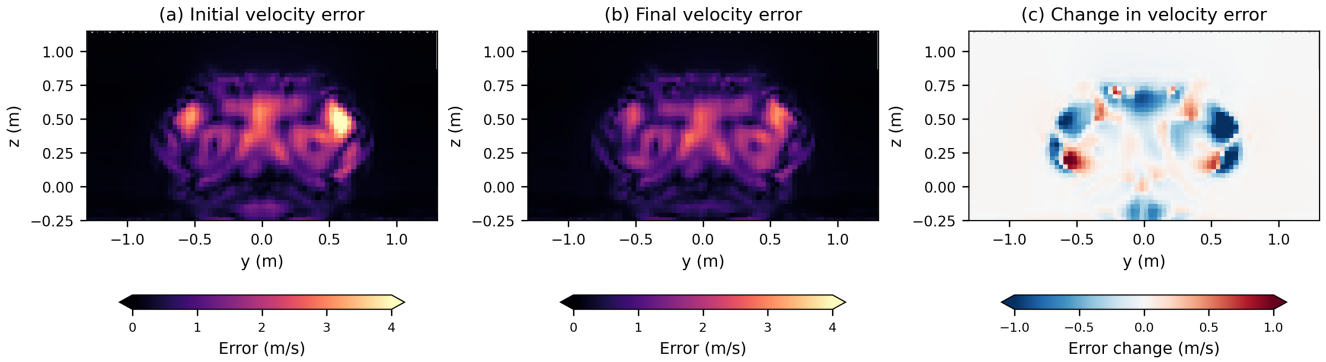


Figure 8: Velocity error at $x = 4.5\,\mathrm{m}$: (a) initial RANS, (b) final EVA and (c) final minus initial.

4.6 Quantitative error reduction

The squared velocity error was integrated over selected volumes of the saved fields:

$$E_{\mathcal{V}} = \int_{\mathcal{V}} |\mathbf{U} - \mathbf{U}_{\mathrm{ref}}|^2 \, \mathrm{d}V. \tag{19}$$

Using the first displayed cycle as the baseline, this measure decreases by 43.27% over the full domain. The regional reductions are:

- **Wake window:** 62.35%, within $4 \le x < 12$, $|y| < 2$ and $z < 2.5$.
- **Roof window:** 56.82%, within $0 < x < 3$, $|y| < 1$ and $0.8 < z < 1.7$.

All coordinates are in metres.

4.7 Eddy-viscosity adaptation

Figure 9 shows how EVA changes the eddy viscosity across the wake. The correction is not uniform: ν_t decreases in parts of the two lateral regions and increases between them and towards the upper wake.

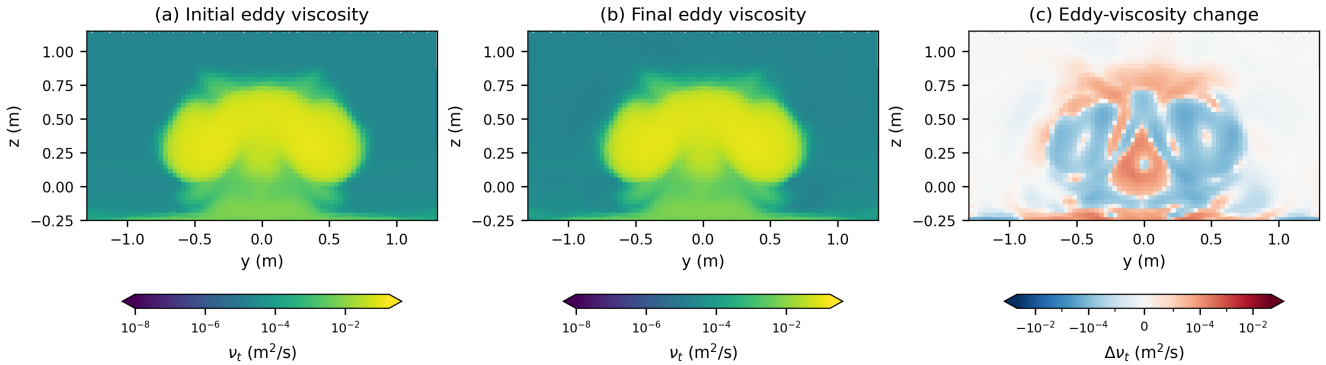


Figure 9: Eddy viscosity in a transverse wake section: (a) initial field, (b) final field and (c) their difference. Logarithmic colour scales are used to make both small and large values visible.

Figure 10 focuses on the lower rear edge, where the final mean adjoint magnitude reaches its maximum. The adjoint field, the viscosity correction and the resulting change in velocity error are displayed. It becomes clear now that an increase in ν_t does not necessarily mean a reduction in velocity error at the same position, since its effect also spreads through the surrounding flow.

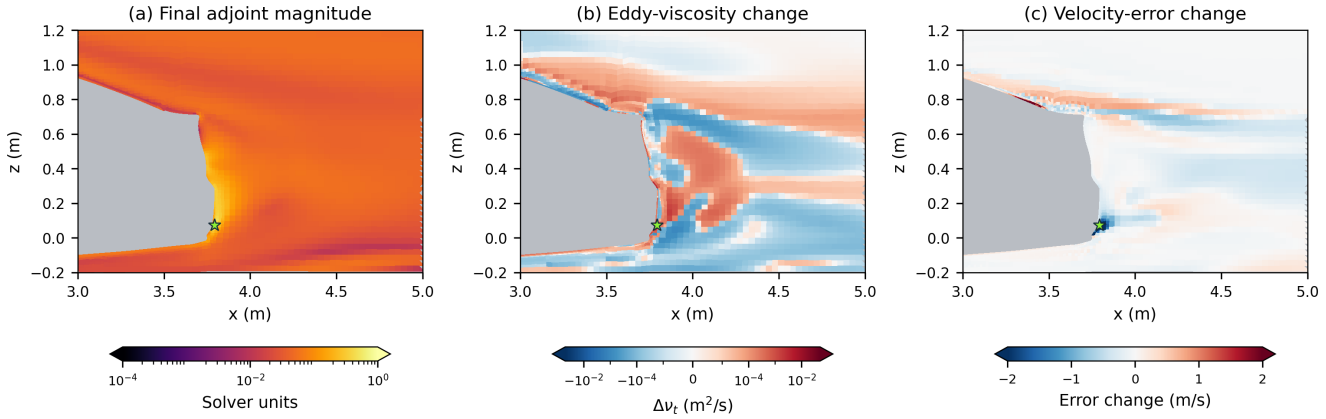


Figure 10: Section near the rear of the vehicle: (a) final mean adjoint magnitude, (b) change in eddy viscosity and (c) change in velocity error. The star marks the maximum-adjoint cell.

4.8 Assessment

EVA improved the agreement between the RANS velocity field and the DES reference by adapting the eddy viscosity. The reduction in velocity error is visible in the wake and around the roof, while the vector comparisons show that the method can improve the flow direction as well as the velocity magnitude. The adaptation produces a spatially varying correction, with both increases and decreases in eddy viscosity. Overall, the results support the use of EVA to improve the RANS prediction for this configuration. At optimisation cycle 38, the final drag coefficient is $C_D = 0.1305$. Drag was not included in the objective function. Further work could test a spatial mask that keeps ν_t fixed in selected regions near the body, while allowing adaptation in the wake. Alternatively, a smoothness penalty on the viscosity correction could be tested to limit abrupt local changes.

References

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